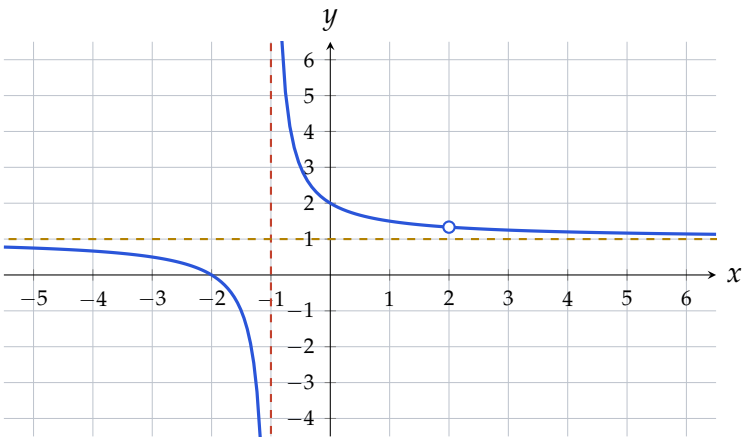
Lesson 1-3 • Limits to Infinity and Horizontal Asymptotes

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: (1) find vertical asymptotes and holes; (2) use the top/bottom rules for limits to infinity and horizontal asymptotes.

The graph	 Graph of $f(x) = \dfrac{x^2-4}{x^2-x-2}$
Vertical asymptotes (VA)	Factor: $f(x) = \dfrac{x^2-4}{x^2-x-2} =$ _____ **Vertical asymptote**: the _____ is 0 after cancelling: _____. **Hole**: a factor that _____: _____, at height $\frac{2+2}{2+1} =$ ____. A hole is a _____ discontinuity. Near a VA the limit is _____: $\lim_{x\to -1^+} f(x) =$ _____, $\lim_{x\to -1^-} f(x) =$ _____.
Horizontal asymptotes (HA)	A horizontal asymptote comes from a limit to _____: $\lim_{x\to\infty} f(x) =$ ____, so the HA is _____. Fun fact: for a rational function with a HA, _____ : the same HA on both ends.
Limits to infinity: top/bottom rules	Compare the _____ (degrees) of the top and the bottom. 1) $\lim_{x\to\pm\infty} \frac{\text{BIG}}{\text{small}} =$ _____ bigger degree on top 2) $\lim_{x\to\pm\infty} \frac{\text{small}}{\text{BIG}} =$ _____ bigger degree on the bottom 3) $\lim_{x\to\pm\infty} \frac{\text{same}}{\text{same}} =$ _____ same degree

Examples	1) $\lim_{x\to\infty} \frac{4x^3 - x}{5 - 2x^2} =$ _____ 2) $\lim_{x\to-\infty} \frac{3x}{x^2 + 7} =$ ____ 3) $\lim_{x\to\infty} \frac{6x^2 + 1}{3 - 2x^2} =$ _____
Summary	